

1. The USC football team wins any one game with probability p , and loses it with probability $1 - p$. Its performance in each game is independent of its performance in other games. Let L_1 be the number of losses before its first win, and let L_2 be the number of losses after its first win and before its second win. Find the joint PMF of L_1 and L_2 .

$$L_1 \sim \text{Poi}_{\text{Geo}}(p)$$

$$P_{L_1}(x) = p \cdot (1-p)^x$$

$$L_2 \sim \text{Poi}_{\text{Geo}}(p)$$

$$P_{L_2}(x) = p \cdot (1-p)^x$$

joint pmf $P_{L_1, L_2}(l_1, l_2) = P_{L_1}(l_1) P_{L_2}(l_2) = p^2 (1-p)^{l_1 + l_2}$

2. Answer the following questions about the Geometric random variable:

- Find the survival function of the $X \sim \text{Geo}_0(p)$, $S_X(n) = \mathbb{P}(X > n)$, $n \in \mathbb{Z}^{\geq 0}$.
- Let $X_1, \dots, X_n$ be i.i.d. $\text{Geo}_0(p)$ random variables. Assume that $Y = \min(X_1, \dots, X_n)$. Find $S_Y(t)$ for $t \in \mathbb{Z}^{\geq 0}$.
- Let $X_1, \dots, X_n$ be i.i.d. $\text{Geo}_0(p)$ random variables. Find the probability that none of them exceeds 5.
- Let $X_1, \dots, X_6$ be i.i.d. $\text{Geo}_0(p)$ random variables. Find the probability that exactly three of them exceeds 5.

$$a) \quad S_X(n) = P(X > n) = \sum_{x=n+1}^{\infty} P_X(x) = \sum_{x=n+1}^{\infty} p(1-p)^x = p \cdot \frac{(1-p)^{n+1}}{1-(1-p)} = (1-p)^{n+1}.$$

$$b) \quad Y = \min(X_1, \dots, X_n)$$

$$\begin{aligned} S_Y(t) &= P(Y > t) = P(\min(X_1, X_2, \dots, X_n) > t) = P(X_1 > t, X_2 > t, \dots, X_n > t) \\ &= P(X_1 > t) P(X_2 > t) \dots P(X_n > t) = \prod_{k=1}^n (1-p)^{t+1} = (1-p)^{(t+1)n}. \end{aligned}$$

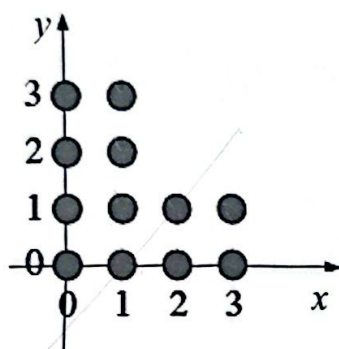
$$\begin{aligned} c) \quad P(\text{none of them exceeds 5}) &= 1 - P(\text{all of them exceeds 5}) \\ &= 1 - S_Y(5) = (1-p)^{6n}. \end{aligned}$$

$$d) \quad Y \sim \text{Bin}(6, (1-p)^6)$$

$$P(Y=3) = \frac{6!}{3!3!} ((1-p)^6)^3 (1-(1-p)^6)^3$$

3. The discrete random variables X and Y have joint pmf $p_{X,Y}(x,y)$ that is nonzero only for the integer values of x and y in the region shown in the figure. Furthermore, the pmf is equal to a constant c for these values of x and y .

- Determine the constant c and $\mathbb{P}(Y \geq X - 1)$
- Determine the marginal pmfs of X and Y . Are they independent? Justify your answer.
- Determine $\text{Cov}(X, Y)$.



$$a) \quad p_{X,Y}(x,y) = c \quad \forall (x,y) \in G$$

$$\sum_{(x,y) \in G} p_{X,Y}(x,y) = 12 \cdot c = 1$$

$$c = 1/12$$

$$P(Y \geq X - 1) = 9 \cdot c = 3/4$$

$$b) \quad P(X=0) = P(X=1) = 4/12 = 1/3$$

$$P(X=2) = P(X=3) = 2/12 = 1/6$$

$$P(Y=0) = P(Y=1) = 4/12 = 1/3$$

$$P(Y=2) = P(Y=3) = 2/12 = 1/6$$

$$P(X=x, Y=y) = \frac{1}{12} \quad \forall (x,y) \in G$$

$$\forall (x,y) \in G, \quad P(X=x, Y=y) \neq P(X=x)P(Y=y)$$

not independent

$$c) \quad \text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$P(XY=0) = 7/12$$

$$P(XY=1) = 1/12$$

$$P(XY=2) = 2/12$$

$$P(XY=3) = 2/12$$

$$\therefore E[XY] = \frac{1}{12} + \frac{1}{12} + \frac{2}{12} + \frac{2}{12} = \frac{11}{12}$$

$$E[X] = E[Y] = 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} = \frac{2}{3} + \frac{1}{2} = \frac{4+3}{6} = \frac{7}{6}$$

$$\therefore \text{Cov}(X, Y) = \frac{11}{12} - \frac{49}{36} = \frac{-16}{36} = -\frac{4}{9}$$

4. Assume that $N_1 \sim \text{Geo}_1(p)$ and $N_2 \sim \text{Pois}(\lambda)$, and let $X_i \sim \text{NegB}(k, r)$ be a sequence of i.i.d random variables when $i \in \mathbb{Z}$, and assume that N_1 and N_2 and X_i 's are independent from one another. Define

$$W = \sum_{i=-N_1}^{N_2-1} X_i$$

Find $\mathbb{E}\{W\}$.

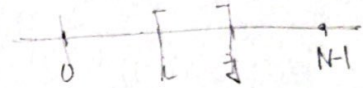
Hint: A negative binomial random variable $\text{NegB}(k, r)$ with probability of success r can be seen as the sum of k independent $\text{Geo}_1(r)$ random variables.

$$\begin{aligned} \mathbb{E}[W] &= \mathbb{E}\left[\mathbb{E}\left[\sum_{i=-N_1}^{N_2-1} X_i \mid N_1, N_2\right]\right] = \mathbb{E}\left[\sum_{i=-N_1}^{N_2-1} \mathbb{E}[X_i \mid N_1, N_2]\right] \\ &= \mathbb{E}\left[\sum_{i=-N_1}^{N_2-1} \mathbb{E}[X_i]\right] = \mathbb{E}[(N_1 + N_2) \mathbb{E}[X_i]] \\ &= \mathbb{E}[X_i] (\mathbb{E}[N_1] + \mathbb{E}[N_2]) \\ &= \frac{k}{p} \cdot \left(\frac{1}{p} + \lambda\right) \end{aligned}$$

5. Let's define two random variables X and Y using an auxiliary random variable Z . Let $Z \sim \text{dU}[0, N-1]$ and $i, j \in \mathbb{N}$ such that $0 < i < j-1 < N-1$. Let

$$X = \begin{cases} 1 & \text{if } 0 < Z < j; \\ 0 & \text{otherwise.} \end{cases}$$

$$Y = \begin{cases} 1 & \text{if } i < Z < N; \\ 0 & \text{otherwise.} \end{cases}$$



Find the covariance between X and Y .

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$= P(X=1) - P(X=1)P(Y=1)$$

$$XY = \begin{matrix} 0 \\ 1 \end{matrix}$$

$$i < Z < j \Rightarrow (j-i-1)$$

$$= \frac{j-i-1}{N} - \frac{j-1}{N} \cdot \frac{N-i-1}{N}$$